

SHRI SANJAYKUMAR V

Software Engineering Intern Candidate · M.Tech Integrated Software Engineering · CGPA 9.12 / 10.0
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EDUCATION

Vellore Institute of Technology — M.Tech Integrated, Software Engineering

Aug 2023 – Present

CGPA: 9.12 / 10.0 (Penultimate Year) | Expected Graduation: 2028

- Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Software Engineering, Computer Networks, Operating Systems, Database Management Systems, Cloud Computing, Machine Learning, Statistics & Probability

TECHNICAL SKILLS

Languages: Python, C++, C, JavaScript, TypeScript, SQL, PHP

CS Fundamentals: Data Structures, Algorithms, Graph Algorithms, OOP, Design Patterns, Concurrency, Multithreading, Synchronization

Software Engineering: System Design, Software Architecture, REST APIs, Microservices, CI/CD, Agile/Scrum, Git/GitHub, Debugging, Testing

Backend & Frontend: Node.js, Express.js, React, Next.js, Firebase, Tailwind CSS, Chart.js

Cloud & DevOps: AWS EC2, Microsoft Azure, Docker, Docker Compose, Linux (Ubuntu), Jenkins, SSH, IaaS

Databases: MongoDB, SQL, Firebase Realtime Database

AI / ML: scikit-learn, XGBoost, LSTM, Random Forest, Gradient Boosting, OpenCV, MediaPipe, RAG, Vector Embeddings, Semantic Search, OpenAI API, Pandas, NumPy, Matplotlib

ENGINEERING PROJECTS

FinTrack — AI-Powered Personal Finance & Group Settlement Platform · Python · React · Node.js · Firebase · REST APIs · Graph Algorithms · OpenAI API 2024

- Architected a full-stack application with a RESTful API layer and real-time data synchronisation across 5+ decoupled modules; applied fault-isolation design patterns to ensure scalability and maintainability.
- Optimized the graph-based debt-settlement algorithm, reducing computational complexity and enabling efficient settlement for multi-user transactions..
- Integrated OpenAI API for AI-driven spending-pattern analysis with natural-language insights; managed full SDLC via Agile/Scrum — sprint planning, Git/GitHub version control, UI/UX wireframing, and integration testing.

Campus AI — Personalised University Assistant (RAG Architecture) · Python · Next.js · React · RAG · Vector Embeddings · OpenAI API · Semantic Search 2024

- Designed and implemented a Retrieval-Augmented Generation (RAG) information-retrieval system with vector embeddings and semantic search, delivering source-cited responses from university documents via intelligent query routing.
- Optimised the end-to-end pipeline — semantic search, live web fallback, and context-aware generation — achieving ~3-second average latency; applied NLP pipeline design and prompt engineering for production-quality AI output.

Solar Energy Production Forecasting — End-to-End ML Pipeline · Python · scikit-learn · XGBoost · LSTM · Gradient Boosting · NASA POWER API 2024

- Designed a complete ML pipeline forecasting AC/DC solar output; trained and benchmarked 5 models (Linear Regression, Random Forest, XGBoost, Gradient Boosting, LSTM) with systematic model selection using MAE and RMSE metrics.
- Engineered multivariate features from the NASA POWER API (irradiance, temperature, humidity); applied LSTM sequential modelling for temporal-dependency capture and automated 7-day forecast generation with visualisation dashboards.

Cloud Infrastructure Deployment & Containerisation · Docker · Docker Compose · AWS EC2 · Microsoft Azure · Linux · Microservices · CI/CD 2023

- Containerised a MERN-stack real-time chat application using Docker Compose, decomposing frontend, backend, and MongoDB into isolated microservices with a CI/CD-ready pipeline — demonstrating distributed-system decomposition and DevOps discipline.
- Provisioned and hardened AWS EC2 and Microsoft Azure VMs; configured security groups, SSH key-pair authentication, NSG rules, and IaaS best practices for secure, production-grade cloud infrastructure.

Vision-Based Finger-Gesture Control System · Python · OpenCV · MediaPipe · Raspberry Pi · TCP/IP · Concurrency · Embedded Systems
2024

- Deployed real-time hand-gesture recognition on Raspberry Pi using OpenCV and MediaPipe; achieved 93–96% classification accuracy via a custom deque-based majority-voting stabilisation algorithm — demonstrating applied ML optimisation and embedded-systems deployment.
- Implemented a low-latency TCP/IP command-transmission pipeline (120–180 ms end-to-end) with application-aware gesture-to-keyboard mapping — showcasing networking fundamentals, concurrency, and systems-level engineering.

Forensica Lite — Lightweight File Forensic Analysis Tool · Python · YARA · Shannon Entropy · SHA-256 · React · Chart.js2023

- Engineered a portable forensic analysis tool combining statistical analysis (Shannon entropy), YARA rule-based pattern detection, SHA-256 hashing, and dynamic risk scoring for rapid incident response — demonstrating security-conscious software design.
- Built an offline React/Chart.js web visualiser rendering locally processed JSON forensic reports with zero backend data transmission, ensuring full data-integrity compliance and demonstrating privacy-by-design engineering principles.

LEADERSHIP & ACTIVITIES

Vice Chairperson & Core Member — VIT Health Club2025 – Present

- Directed end-to-end execution of the "Hack the Hospital" Hackathon (200+ participants), leading cross-functional team coordination, sprint planning, and technical workshop delivery under deadline pressure — demonstrating ownership and engineering culture-building.
- Applied Agile project-management principles across multiple simultaneous workstreams, reflecting the collaborative, execution-focused engineering mindset aligned with Google's team culture.

Sponsorship Lead — Rivera & Gravitas Tech Festivals, VIT2025 – 2026

- Secured INR 60+ lakhs in non-monetary sponsorships through structured negotiations with VCare, Joy Innovations, and Jaisha; managed multi-stakeholder deliverables concurrently for two simultaneous 1,000+ attendee university tech festivals.